

Programming Assignment 3 (Deep Reinforcement Learning)

In this assignment we will implement Deep Reinforcement Learning.

Consider the problem of training a Deep Q-learning Network (DQN) for a vacuum cleaner in a grid environment. Our environment is an $m \times n$ grid in which there are $p < m \cdot n$ dirty squares. Target m, n and p values are $m, n = 15, p = 10$. We have a vacuum cleaner starting at a corner position $(0, 0)$ that needs to clean the given dirty squares in as short of a path as possible. The vacuum cleaner can move only in the directions of N, S, E, W and it has to collect all the trash and the job is done when the last dirty tile has been cleaned.

To implement Deep Q-learning you can use the Breakout DQN architecture or Actor-Critic architecture that we have discussed in class.

a) Create a framework to visualize and simulate the given problem. For visualizing the results you can use text mode or you can use pygame library as well. Consider a fixed smaller environment initially for testing your algorithm (fixed dimensions and fixed positions for the dirty tiles), then increase the sizes.

b) Define the reward function for the given problem.

c) Construct a DQN to solve the problem.

d) Train the DQN and adjust the reward function and system parameters as necessary.

e) Visualize reward vs. learning epochs, as well as number of steps that the robot needs in order to complete the task.

f) Explore the problem further by changing dimensions, number of dirty tiles and considering random configurations for the problem.

Rubric: Agent can solve a 5x5 fixed environment 60%

Agent can solve a 10x10 fixed environment 80%

Agent can solve a 15x15 fixed environment 95%

Use a pygame visualization: +5%